

JUGAL PARMESHWAR PAKHARE

AI/ML Engineer | MLOps | Software Engineering

Jalna, Maharashtra, India | jugalpkhr@gmail.com

<https://portfoliojugal.vercel.app/> | <https://github.com/jugal4096> | <https://www.linkedin.com/in/jugal-pakhare-a58653359>

PROFESSIONAL SUMMARY

B.Tech Information Technology student focused on **Machine Learning, MLOps, and Software Engineering**, with hands-on experience building data-driven applications and ML systems. Skilled in **Python, C++, SQL, Scikit-learn, Pandas, NumPy, Git/GitHub, REST APIs, Docker, and web technologies**. Experienced in data preprocessing, feature engineering, model training, evaluation, application development, and deployment. Currently developing expertise in **ML pipelines, experiment tracking, data versioning, CI/CD, model deployment, monitoring, and cloud infrastructure**. Interested in building reliable, scalable, and production-ready AI/ML systems.

EDUCATION

Government College of Engineering, Chhatrapati Sambhajinagar (GECA) <i>B.Tech. in Information Technology</i>	2024 – 2027 <i>Chhatrapati Sambhajinagar District, Maharashtra</i>
Government Polytechnic, Jalna <i>Diploma in Civil Engineering</i>	2018 – 2023 <i>Jalna, Maharashtra</i>
Maharashtra SSC Board <i>Secondary School Certificate (SSC) Percentage: XX%</i>	2017 – 2018 <i>Jalna, Maharashtra</i>

EXPERIENCE

Larsen & Toubro (L&T) <i>Diploma Engineering Trainee / Junior Engineer Trainee</i>	Oct 2023 – Jul 2024 <i>Meerut, Uttar Pradesh</i>
- Worked on the Delhi–Meerut Regional Rapid Transit System (RRTS), gaining experience in large-scale engineering project execution, technical coordination, documentation, and process adherence. - Worked in a structured project environment requiring attention to safety, quality, documentation, coordination, and disciplined execution of engineering activities.	

SELECTED PROJECTS

PopulationVision – AI Population Forecasting System <i>Python, Pandas, NumPy, Scikit-learn, Machine Learning, Data Analysis</i>	GitHub
- Built an AI/ML-based population forecasting system for India using historical demographic data and machine learning workflows. - Implemented data preprocessing, feature engineering, model training, evaluation, prediction, and research-oriented demographic analytics. - Designed the workflow around an ML lifecycle that can be extended with model versioning, automated pipelines, deployment, and monitoring.	
RailAI – Railway Intelligence and Delay Prediction <i>Python, Machine Learning, Data Analysis, Rule-Based Systems</i>	GitHub
- Developed a hybrid railway intelligence system for the Jalna–Chhatrapati Sambhajinagar corridor to analyze train movement, delays, and crossing conditions. - Combined rule-based railway operating logic with machine learning for train crossing and delay prediction. - Structured the system into data processing, operational logic, prediction, and application components for maintainability.	
Task Tracker – MERN Full-Stack Application <i>MongoDB, Express.js, React, Node.js, REST APIs</i>	GitHub
- Built and deployed a full-stack task management application using MongoDB, Express.js, React, and Node.js. - Implemented RESTful APIs, CRUD operations, database integration, frontend–backend communication, and production deployment.	
GECA Student Academic Planner <i>JavaScript, Firebase, Firestore, HTML, CSS, Capacitor</i>	GitHub
- Developed an academic platform with SGPA/CGPA calculation, dashboard functionality, authentication, and persistent student data. - Implemented guest and authenticated workflows using browser storage, Firebase Authentication, and Firestore.	

TECHNICAL SKILLS

Programming: Python, C++, JavaScript, SQL
Machine Learning: Machine Learning, Scikit-learn, Supervised Learning, Regression, Classification, Feature Engineering, Model Evaluation, Predictive Modeling, Data Preprocessing, Time-Series Forecasting
Data & Analytics: Pandas, NumPy, Exploratory Data Analysis, Data Cleaning, Data Transformation, Data Visualization, Statistical Analysis, SQL
MLOps: ML Pipelines, Model Serialization, MLflow, DVC, Docker, CI/CD, Experiment Tracking, Data Versioning, Model Deployment, Model Monitoring, Logging, REST APIs, FastAPI
Software Engineering: Object-Oriented Programming, Data Structures, Algorithms, RESTful APIs, Backend Development, API Integration, Testing, Debugging, Version Control, Software Design
Web Development: React, Node.js, Express.js, JavaScript, HTML, CSS, Firebase
Databases: MongoDB, SQL, Firebase Firestore

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Linux, Virtual Environments, pip, Netlify, Vercel

Cloud: AWS, Google Cloud, Microsoft Azure – fundamentals

MLOPS FOCUS

ML Lifecycle: Data Ingestion → Data Validation → Preprocessing → Feature Engineering → Training → Evaluation → Experiment Tracking → Versioning → Deployment → Monitoring.

Current Focus: Productionizing ML systems, reproducible experiments, automated ML pipelines, containerization, CI/CD automation, model serving, monitoring, observability, and cloud-based ML deployment.

ADDITIONAL

Career Focus: Machine Learning Engineering, MLOps Engineering, AI Engineering, Software Engineering, ML Infrastructure

Engineering Interests: Scalable ML Systems, Model Deployment, Data Pipelines, AI Infrastructure, Automation, Reliable Software Systems